

RAHUL R – 192412255

ECA0502 EMF

EXP 2 TESTCASE 2

Transformation between Cartesian, Cylindrical and spherical coordinate system

PROGRAM:

```
EMF EX1 (ii).sci (C:\Users\rahu\OneDrive\Documents\EMF EX1 (ii).sci) - SciNotes
File Edit Format Options Window Execute ?
[Icons]
EMF EX1 (ii).sci (C:\Users\rahu\OneDrive\Documents\EMF EX1 (ii).sci) - SciNotes
EMF EX1.sci X EMF EX1 (ii).sci X EMF EXP1 (iii).sci X
1 clc;
2 r=input('Enter the value of r=');
3 pi=input('Enter the value of pi=');
4 z=input('Enter the value of z=');
5 x=r*cosd(pi);
6 y=r*sind(pi);
7 Z=z;
8 disp('cylindrical to cartesian coordinate system of p(x,y,z)=[x y z]');
9 r=sqrt(r^2+z^2);
10 pi=pi;
11 theta=acosd(z/r);
12 disp('cylindrical to spherical coordinate system of s(r,pi,theta)=[r pi theta]');
13
```

OUTPUT:

Scilab 2026 1.0 Console ? ? X

Enter the value of r=4
Enter the value of pi=60
Enter the value of z=-3

"cylindrical to cartesian coordinate system of p(x,y,z)="
2. 3.4641016 -3.
"cylindrical to spherical coordinate system of s(r,pi,theta)="
5. 60. 126.8699
exec: Wrong number of output argument(s): 0 expected.
-->

Variable Browser ? ? X

Name	Value	Type	Visibility	Memory
pi	71.6	Double	local	216 B
r	4.36	Double	local	216 B
theta	133	Double	local	216 B
x	1	Double	local	216 B
y	3	Double	local	216 B
z	-3	Double	local	216 B

Command History ? ? X

60
-3
4
-25
100
4
60
-3
1
3
-3
4
60
-3

OBSERVATION:

Test case (10, 11)

$$\rho = 4$$

$$\phi = 60^\circ$$

$$z = -3$$

cyl $\rightarrow$ cartesian coordinates

$$x = r \cos \phi$$

$$= 4(0.5) = 2$$

$$\Rightarrow \boxed{x = 2}$$

$$y = r \sin \phi$$

$$= 4(0.866025)$$

$$\boxed{y = 3.4641}$$

$$\boxed{z = -3}$$

cyl $\rightarrow$ spherical coordinates

$$r = \sqrt{\rho^2 + z^2}$$

$$= \sqrt{4^2 + (-3)^2} = \sqrt{25}$$

$$\boxed{r = 5}$$

$$\boxed{\phi = 60^\circ}$$

$$\theta = \cos^{-1}\left(\frac{z}{r}\right)$$

$$= \cos^{-1}\left(\frac{-3}{5}\right)$$

$$\boxed{\theta = 126.870}$$

$$(x, y, z) = (2, 3.4641, -3)$$

$$(r, \phi, \theta) = (5, 60, 126.870)$$